§ Lecture 18.0

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Claim 5 $\lim_{n\to\infty} a_n = c \iff \limsup a_n = c = \liminf a_n.$

Proof: $\Longrightarrow$ Previous class.

$\Longleftarrow$ Assume $\limsup a_n = c = \liminf a_n$

Fix $\varepsilon > 0$.

Our goal is to show that $\exists N \geq 1$ s.t.

$$c-\varepsilon \leq a_n \leq c+\varepsilon \quad \forall n \geq N.$$

Note that ①

$$c-\varepsilon < c = \sup_{N\geq 1}(a_N^-)$$

$\Rightarrow$ $c-\varepsilon$ can't be an upper bound (as sup is the smallest upper bound)

$\Rightarrow$ $\exists N_1 \geq 1$ s.t. $a_{N_1}^- > c-\varepsilon$

Then $\forall n \geq N_1$ $\quad a_n \geq \inf_{n\geq N_1} a_n = a_{N_1}^- > c-\varepsilon$

$\Rightarrow$ $\forall \varepsilon > 0$ $\exists N_1 \geq 1$ s.t. $\forall n \geq N_1$

$$a_n > c-\varepsilon. \quad \rightarrow ①$$

② $\quad c+\varepsilon > c = \inf_{N\geq 1}(a_N^+)$

Since $\inf_{N\geq 1}(a_N^+)$ is the greatest lower bound on $\{a_N^+ : N \geq 1\}$, $c+\varepsilon$ can't be a lower bound.

$\Rightarrow$ $\exists N_2 \geq 1$ s.t. $a_{N_2}^+ < c+\varepsilon$

Then $\forall n \geq N_2$ $\quad a_n \leq \sup_{n\geq N_2} a_n = a_{N_2}^+ < c+\varepsilon.$

Thus $\forall \varepsilon > 0$, $\exists N_2 \geq 1$ s.t. $a_n < c+\varepsilon$ $\forall n \geq N_2.$ $\rightarrow ②$

Combining ① & ② we get

$\forall \varepsilon > 0$ $\exists N = \max\{N_1, N_2\} \geq 1$ s.t.

$$|a_n - c| < \varepsilon \quad \forall n \geq N.$$

$$\Rightarrow \lim_{n\to\infty} a_n = c.$$

Lemma (Comparison principle): Let $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ be two sequences of reals with $a_n \leq b_n$ $\forall n \geq 1$. Then

① $\inf_{n\geq 1}(a_n) \leq \inf_{n\geq 1}(b_n)$ ② $\sup_{n\geq 1}(a_n) \leq \sup_{n\geq 1}(b_n)$

③ $\liminf_{n\to\infty}(a_n) \leq \liminf_{n\to\infty}(b_n)$ ④ $\limsup_{n\to\infty}(a_n) \leq \limsup_{n\to\infty}(b_n).$

Proof: ① $b_n \geq a_n \geq \inf_{n\geq 1}(a_n)$ $\forall n \geq 1$

$\Rightarrow$ $\inf(a_n)$ is a lower bound on (b_n). Then since all lower bounds are smaller than $\inf(b_n)$, we have

$$\inf(a_n) \leq \inf(b_n).$$

② $\quad a_n \leq b_n \leq \sup(b_n)$

$\Rightarrow$ $\sup(b_n)$ is an upper bound on (a_n), this means

$$\sup(a_n) \leq \sup(b_n).$$

③ Fix N. Then

$$a_n \leq b_n \quad \forall n \geq N$$

$$\Rightarrow \sup_{n\geq N}(a_n) \leq \sup_{n\geq N}(b_n)$$

$$\Rightarrow a_N^+ \leq b_N^+ \quad \forall N \geq 1$$

$$\Rightarrow \inf(a_N^+) \leq \inf(b_N^+)$$

That is $\limsup(a_n) \leq \limsup(b_n).$

Squeeze test: Let $(a_n)_{n\geq 1}$, $(b_n)_{n\geq 1}$, $(c_n)_{n\geq 1}$ be sequences of real numbers such that

① $a_n \leq b_n \leq c_n$ $\forall n \geq 1.$ ② $\lim_{n\to\infty} a_n = l = \lim_{n\to\infty} c_n$

Then $\lim_{n\to\infty} b_n = l.$

Proof 1. Applying composition principle as well as the fact that

$$\lim_{n\to\infty} b_n = l \iff \limsup_{n\to\infty} b_n = l = \liminf_{n\to\infty} b_n.$$

We have $\quad a_n \leq b_n \leq c_n \Rightarrow \limsup a_n \leq \limsup b_n \leq \limsup c_n$

$$\Rightarrow l \leq \limsup b_n \leq l$$

$$\Rightarrow \limsup b_n = l.$$

Similarly, $\liminf b_n = l$

Thus $\lim_{n\to\infty} b_n = l.$

Proof 2: Since $\lim_{n\to\infty} a_n = l$

$\Rightarrow$ $\forall \varepsilon > 0$ $\exists N_1 \geq 1$ s.t. $|a_n - l| \leq \varepsilon$ $\forall n \geq N_1.$ $\rightarrow ①$

or $\quad l-\varepsilon \leq a_n \leq l+\varepsilon$

Since $\lim_{n\to\infty} c_n = l$

$\Rightarrow$ $\forall \varepsilon > 0$ $\exists N_2 \geq 1$ s.t. $|c_n - l| \leq \varepsilon$ $\forall n \geq N_2$ $\rightarrow ②$

or $\quad l-\varepsilon \leq c_n \leq l+\varepsilon.$

$\Rightarrow$ $\quad l-\varepsilon \leq a_n \leq b_n \leq c_n \leq l+\varepsilon$ $\quad \forall n \geq \max\{N_1, N_2\}$

$$\Rightarrow \lim_{n\to\infty} b_n = l.$$

Proof 3: Fails

$$a_n \leq b_n \leq c_n$$

$$\lim_{n\to\infty} a_n \leq \lim_{n\to\infty} b_n \leq \lim_{n\to\infty} c_n \qquad \left(\mathrm{LIM}_{n\to\infty}\, a_n = \lim_{n\to\infty} a_n\right).$$

↑ This doesn't go through as we don't know if $\lim_{n\to\infty} b_n$ exists.

Completeness of reals.

Theorem: Every sequence $(a_n)_{n\geq 1}$ of reals is a Cauchy sequence iff it is convergent.

Proof. We have shown already that every convergent sequence is Cauchy. We just need to show that every Cauchy sequence is convergent.

Let $(a_n)_{n\geq 1}$ be a Cauchy sequence.

Then $(a_n)_{n\geq 1}$ is a bounded sequence. That is there is a real number M s.t. $\quad -M \leq a_n \leq M$

From comparison principle $\Rightarrow$ $\limsup(a_n)$ and $\liminf(a_n)$ are both bounded by M, hence finite.

Let $\limsup_{n\to\infty} a_n = L^+$ and $\liminf_{n\to\infty} a_n = L^-.$

Our goal is to show that $L^+ = L^-.$

Let $\varepsilon > 0$ be an arbitrary real number. Since $(a_n)_{n\geq 1}$ is Cauchy $\exists N \geq 1$ s.t. $\quad a_N - \varepsilon \leq a_n \leq a_N + \varepsilon$ $\quad \forall n \geq N.$

Then $\quad a_N - \varepsilon \leq a_n$ $\quad \forall n \geq N \Rightarrow$

$$\begin{aligned} a_N - \varepsilon &\leq \inf_{n\geq N}(a_n) \\ &= a_N^- \\ &\leq \sup_{N\geq 1} a_N^- = L^- \end{aligned}$$

ie. $\quad a_N - \varepsilon \leq L^-.$ $\rightarrow ①$

Similarly, $\quad a_N + \varepsilon \geq a_n$ $\forall n \geq N \Rightarrow$

$$\begin{aligned} a_N + \varepsilon &\geq \sup_{n\geq N}(a_n) \\ &= a_N^+ \\ &\geq \inf_{N\geq} a_N^+ = L^+ \end{aligned}$$

$\Rightarrow$ $\quad L^+ \leq a_N + \varepsilon.$ $\rightarrow ②$

From ① & ② $\quad L^+ - L^- \leq 2\varepsilon.$

Further $L^+ \geq L^-$, we have $\quad 0 \leq L^+ - L^- \leq 2\varepsilon$ $\rightarrow ③$

$\Rightarrow$ $\forall \varepsilon > 0$, $\quad |L^+ - L^-| \leq 2\varepsilon \iff L^+ = L^-.$

Remark: ① $|x-a| \leq \varepsilon$ $\forall \varepsilon > 0 \Rightarrow x = a.$

Suppose $x \neq a$, ie. $|x-a| > 0$. Choose $\varepsilon = \frac{|x-a|}{2}$

Then $\quad |x-a| \leq \frac{|x-a|}{2}$ or $\quad 1 \leq \frac{1}{2}.$ [Contradiction].

② $\forall \varepsilon > 0$ $\exists N \geq 1$ s.t. $\quad |a_n - L| \leq \varepsilon$ $\quad \forall n \geq N.$

$$\overset{\times}{\Rightarrow} a_n = L \quad \forall n \geq N., \text{ why?}$$

(Because N depends on ε)

§ Lecture 18.1

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An example of limit:

Claim: $\lim_{n\to\infty} n^{-q} = 0 \quad \forall$ rational $q>0$.

Proof: Since $q>0$, write $q = \frac{a}{b}$, where $a, b \in N-\{0\}$.

Note that $n^{a/b} = (n^{1/b})^{a}$

Let $n^{1/b} = p \Rightarrow n = p^{b}$

$$n+1 > n \Rightarrow (n+1)^a > n^a$$
$$\Rightarrow (n+1)^{a/b} > n^{a/b} \qquad [x>y \Leftrightarrow x^{1/k} > y^{1/k} \ \forall k\geq 1]$$
$$\Rightarrow (n+1)^q > n^q$$
$$\Rightarrow (n+1)^{-q} < n^{-q}$$

That n^{-q} is a decreasing function of n for all $q>0$.

Thus sequence $(n^{-q})_{n\geq 1}$ is decreasing and bounded by zero from below. Thus, $\lim_{n\to\infty} n^{-q}$ exists.

Let $l = \lim_{n\to\infty} n^{-q} = \lim_{n\to\infty} n^{-a/b}$

$$\Rightarrow l^b = \lim_{n\to\infty} n^{-a/b \times b}$$
$$= \lim_{n\to\infty} n^{-a}$$
$$= [\lim_{n\to\infty} n^{-1}]^a$$
$$= 0^a$$
$$= 0.$$

$\Rightarrow \boxed{l=0}$.

Subsequences.

Definition: Let $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ be sequences of real numbers. We say that $(b_n)_{n\geq 1}$ is a subsequence of $(a_n)_{n\geq 1}$ iff there exists a function $f: N \to N$ which is strictly increasing s.t.

$$b_n = a_{f(n)} \quad \forall n \in N.$$

[Note f is an injective function otherwise it can't be strictly increasing. It may not be bijective though.]

Example: $(b_n)_{n\geq 1}$, where $b_n = a_{2n}$ is a subsequence of $(a_n)_{n\geq 1}$.

Lemma 1: Let $(a_n)_{n\geq 1}$ be a sequence of reals and $L \in R$. Then following statements are equivalent.

(i) $\lim_{n\to\infty} a_n = L$.

(ii) Every subsequence of $(a_n)_{n\geq 1}$ converges to L.

Proof: (ii) $\Rightarrow$ (i) is trivial as $(a_n)_{n\geq 1}$ is a subsequence of itself. (Choose $f(n)=n$).

Since every subsequence of $(a_n)_{n\geq 1}$ converges to $L \Rightarrow \lim_{n\to\infty} a_n = L$.

(i) $\Rightarrow$ (ii) Given $\lim_{n\to\infty} a_n = L. \Rightarrow \forall \varepsilon>0 \ \exists N\geq 1$ s.t. $|a_n - L| \leq \varepsilon \ \forall n \geq N$.

Let $f: N \to N$ be an arbitrary strictly increasing function.

Then $f(n) \geq n$. $\forall n \in N$.

[By induction: $f(0) \in N \Rightarrow f(0) \geq 0$

Induction step: $f(n) \geq n$. Then

$f(n+1) > f(n)$

$\Rightarrow f(n+1) \geq f(n)+1 \geq n+1$.]

Then

$f(n) \geq n \geq N$, ie.

$|a_{f(n)} - L| \leq \varepsilon$

$\Rightarrow \lim_{n\to\infty} a_{f(n)} = L$ for any strictly increasing function f.

Lemma 2. Let $(a_n)_{n\geq 1}$ be a sequence of reals and $L \in R$. Then following two statements are equivalent.

(i) L is a limit point of $(a_n)_{n\geq 1}$.

(ii) There exists a subsequence of $(a_n)_{n\geq 1}$ that converges to L.

Proof: (ii) $\Rightarrow$ (i) (Easy direction)

Suppose $(a_{f(n)})_{n\geq 1}$ be a subsequence of $(a_n)_{n\geq 1}$ such that

$\lim_{n\to\infty} a_{f(n)} = L$.

$\Rightarrow \forall \varepsilon>0 \ \exists N_0 \geq 1$ s.t. $|a_{f(n)} - L| \leq \varepsilon \ \forall n \geq N_0. \longrightarrow$ ①

Goal: $\forall \varepsilon>0 \ \forall N\geq 1 \ \exists m \geq N$ s.t. $|a_m - L| \leq \varepsilon$.

Fix $\varepsilon>0$. Fix $N \geq 1$. We need to find m s.t.

① $m \geq N$, ② $|a_m - L| \leq \varepsilon$.

Pick $j = \max\{N, N_0\}$

From ① $|a_{f(j)} - L| \leq \varepsilon$.

Call $f(j) = m$. $\Rightarrow$ ① $m = f(j) \geq j \geq N$

② $|a_m - L| \leq \varepsilon$.

$\Rightarrow L$ is a limit point of $(a_n)_{n\geq 1}$.

(i) $\Rightarrow$ (ii) Difficult direction.

Given L is a limit point of $(a_n)_{n\geq 1}$. That is

$\forall \varepsilon>0 \ \forall N\geq 1 \ \exists n \geq N$ s.t. $|a_n - L| \leq \varepsilon. \longrightarrow$ ①

Goal: To construct a subsequence that converge to L.

Recursive construction: $N=1$, $\varepsilon = 1$

$\exists n \geq 1$ s.t. $|a_n - L| \leq 1$.

Call $f(1)$, the smallest n satisfying above.

Recursive step: Let $f(j)$ has been defined.

Apply eqn ① with $\varepsilon = \frac{1}{j+1}$, $N = f(j)+1$

There is $n \geq f(j)+1$ s.t. $|a_n - L| \leq \frac{1}{j+1}$

Let $f(j+1)$ be the smallest such n. [Well defined from well ordering principle]

Thus $f(j)$ is defined for all naturals.

By construction: ① $f(j+1) \geq f(j)+1 > f(j)$

② $|a_{f(j)} - L| \leq \frac{1}{j}$

From Archimedean property choose $J_\varepsilon \geq 1$

Then $\forall j \geq J \quad |a_{f(j)} - L| \leq \frac{1}{J} \leq \varepsilon$

$\Rightarrow \lim_{j\to\infty} a_{f(j)} = L$.

Recall: ① Convergence $\Leftrightarrow$ Cauchy.

② Bounded + monotone $\Rightarrow$ Convergence

③ Bounded $\overset{??}{\Longrightarrow}$ convergence

Maybe weaker it has limit points or only a subsequence converges !!

Theorem (Bolzano–Weierstrass theorem): Every bounded sequence of reals has at least one convergent subsequence. [Compactness]

Proof: Given: A real sequence $(a_n)_{n\geq 1}$ such that $\exists M>0$ and

$M \leq a_n \leq M. \rightarrow$ ①

This means that $\limsup_{n\to\infty} a_n$ and $\liminf_{n\to\infty} a_n$ are real (bounded).

Let $L = \limsup_{n\to\infty} a_n$.

If we can show that L is a limit point, then there is a subsequence that converges to L. [Lemma 2 above].

Limit point: $\forall \varepsilon>0 \ \forall N\geq 1 \ \exists n \geq N$ s.t. $|a_n - L| \leq \varepsilon$.

Fix $\varepsilon>0$. Fix $N \geq 1$.

Note that $L+\varepsilon > L = \inf_{M\geq 1} (a_M^+)$

$\Rightarrow L+\varepsilon$ cannot be a lower bound on $(a_M^+)_{M\geq 1}$.

$\Rightarrow \exists M_0 \geq 1$ s.t. $a_{M_0}^+ < L+\varepsilon$.

That is $\forall n \geq M_0 \quad a_n \leq a_{M_0}^+ < L+\varepsilon. \longrightarrow$ ①

Mind that M_0 is some number independent of N.

Further $L-\varepsilon < L = \inf_{M\geq 1} (a_M^+)$

$\forall M \geq 1 \quad a_M^+ > L-\varepsilon$

$\sup_{n\geq M} a_n > L-\varepsilon$

$\Rightarrow L-\varepsilon$ can't be an upper bound. That is $\exists m_1 \geq M$ s.t.

$a_{m_1} > L-\varepsilon$.

Rewriting: $\exists M_0 \geq 1 \ \forall m \geq M_0 \quad a_m < L+\varepsilon$

$\forall M \geq 1 \ \exists m_1 \geq M \quad a_{m_1} > L-\varepsilon \rightarrow$ ②

Let $P = \max\{M_0, N\}$

$\exists n \geq P \quad a_n > L-\varepsilon \longrightarrow$ ③

$\Rightarrow n \geq P \geq M_0 \quad a_n < L+\varepsilon \rightarrow$ ④

Since $n \geq P \geq N$, we have

$|a_n - L| < \varepsilon$.

Thus $\forall \varepsilon>0$, $\forall N\geq 1 \ \exists n \geq N$ s.t. $|a_n - L| < \varepsilon$.

Thus L is a limit point of $(a_n)_{n\geq 1}$.